Parabola cap sets over $\mathbb{F}_{6561}$: two order-41 coefficient cosets of opposite quadratic character

Abstract

We consider unions of punctured parabolas $\{(x, cx^2) : x \in \mathbb{F}_{6561}^{\times}\}$. For coefficients in the multiplicative subgroup H of order 41, the exact maximum is five, with 328 attaining coefficient sets forming one free scaling–Frobenius orbit. We extend this classification to $D = H \cup gH$, where g is a specified primitive element: the maximum remains five, and the 656 attaining sets are the 328 in each coset. Indeed, every cross-coset coefficient pair already gives a three-point obstruction through one of its two repeated-label discriminants. Each attaining union has 32 800 points in $\mathbb{F}_3^{16}$. The finite part of the H classification is certified by an increasing-extension check of all 929 independent sets containing a fixed vertex. A companion supplement gives an alternate upper certificate and portable checks for both domains.

1 The restricted coefficient problem

A *cap* in an affine space over $\mathbb{F}_3$ is a set containing no three distinct points whose sum is zero. For a finite field F of characteristic 3 and a coefficient set $S \subseteq F^{\times}$, put

$$\mathcal{P}(S) = \{(x, cx^2) : x \in F^{\times},\ c \in S\} \subseteq F^2. \tag{1}$$

The exclusion of $x = 0$ is part of the construction. Distinct coefficients give disjoint punctured parabolas, so

$$|\mathcal{P}(S)| = (|F| - 1)|S|. \tag{2}$$

Grace and Voloch study this construction in [2, Section 2]. Their unnumbered question immediately before Lemma 2.5 asks which numbers and choices of coefficients yield a cap when the extension degree is even. We answer the restriction of that question to the order-41 subgroup of $\mathbb{F}_{3^8}^{\times}$ and to its union with a coset of opposite quadratic character. We determine all optimal coefficient sets in both domains.

Fix the model and elements

$$\begin{aligned} F &= \mathbb{F}_3[z]/(z^8 + z^2 + 2), \\ g &= z^3 + z^2 + 2, \\ h &= g^{160} = z^6 + z^5 + 2z^2 + z, \\ H &= \langle h \rangle, \qquad D = H \cup gH. \end{aligned} \tag{3}$$

Thus $|F| = 6561$, g is primitive, and H has order 41. The elementary finite checks establishing these particular coordinate choices are described in Section 6. Since the multiplicative group of a finite field is cyclic, H is its unique subgroup of order 41.

Theorem 1.1. *For F, h, H as in* (3)*,*

$$\max\{\,|S| : S \subseteq H,\ \mathcal{P}(S) \text{ is a cap}\,\} = 5.$$

The coefficient set

$$S_0 = \{1, h^2, h^6, h^{13}, h^{23}\} \tag{4}$$

attains the maximum, and $|\mathcal{P}(S_0)| = 32\,800$. There are exactly 328 *attaining subsets of H. They are precisely*

$$uS_0^{3^j} = \{uc^{3^j} : c \in S_0\}, \qquad u \in H, \quad 0 \le j < 8, \tag{5}$$

and all 328 *sets in this display are distinct.*

Theorem 1.2. *For D as in* (3)*,*

$$\max\{\,|S| : S \subseteq D,\ \mathcal{P}(S) \text{ is a cap}\,\} = 5.$$

There are exactly 656 *attaining coefficient sets. They are the* 328 *sets in* (5) *and the* 328 *sets*

$$guS_0^{3^j}, \qquad u \in H, \quad 0 \le j < 8.$$

Every such cap has 32 800 *points.*

The maximum and the number of attaining sets are intrinsic to H; the exponents in (4) refer to the specified generator h. The H classification is under the stated action on coefficient subsets. Theorem 1.2 lists two disjoint classes; it does not assert that they form one orbit under an action preserving D. It does not assert a classification of ambient affine equivalences or of all caps in $\mathbb{F}_3^{16}$.

The value 32 800 is an attainment result for these restricted problems. Grace–Voloch's Table 2 gives an unrestricted 20-coefficient construction of 131 200 points in the same dimension, found by random search and marked nonexhaustive. Theorems 1.1 and 1.2 therefore give neither a larger general cap nor an unrestricted coefficient optimum. Their content is the exact maximum and classification in the specified coefficient domains.

Our reduction uses the discriminant appearing in [2, Lemma 2.4]; we give the short specialization needed for an equivalence, including the repeated-coefficient cases. Frobenius symmetry is also used in [2, Lemma 2.5 and Remark 2.6]. These numbers refer to the published version. The associated implementation [1] is an antecedent for the choice of field polynomial and coefficient tests. The finite certificate below concerns the specific subgroup restriction and is supplied with its own verification procedure.

2 The exact coefficient criterion

Call $d \in F$ a square if $d = v^2$ for some $v \in F$; this convention includes zero. Write

$$\Delta(a, b, c) = -(ab + ac + bc). \tag{6}$$

The domain D contains no antipodal pair. Within either coset this follows from the odd order of H. Across cosets it follows because -1 is a square, each member of H is a square, and each member of gH is a nonsquare. In particular, $a + b \neq 0$ also when $a = b$.

Grace–Voloch's displayed question allows $x = 0$, whereas (1) is punctured. This makes no difference to caphood for $S \subseteq D$: a distinct zero-sum triple using $(0, 0)$ would require (x, ax^2) and $(-x, bx^2)$ with $x \neq 0$ and $a+b = 0$, which is impossible in D. Thus our restricted classifications also apply when the common origin is included; the point count then increases by one.

Lemma 2.1 (Normalized equation)**.** *Let $a, b, c \in D$. A zero-sum triple with these coefficient labels can be normalized to the three points*

$$(t, at^2), \qquad (1, b), \qquad (-t - 1, c(t + 1)^2), \tag{7}$$

where $t \neq 0, -1$, and its second-coordinate equation is

$$(a + c)t^2 + 2ct + (b + c) = 0. \tag{8}$$

Every root of this quadratic is admissible, and a root exists if and only if $\Delta(a, b, c)$ is a square. A zero-sum triple in (7) *is repeated if and only if $a = b = c$ and $t = 1$.*

Proof. Write the original nonzero first coordinates as x_1, x_2, x_3, with $x_1 + x_2 + x_3 = 0$. Applying the invertible map $(x, y) \mapsto (x/x_2, y/x_2^2)$ preserves the coefficient labels and the zero-sum condition, and gives first coordinates $t, 1, -t - 1$, where $t = x_1/x_2$. This proves (8), as well as its converse. The coefficient $a + c$ is nonzero. The polynomial's values at 0 and -1 are $b + c$ and $a + b$, respectively, so neither excluded value is a root. Its discriminant in characteristic 3 is

$$(2c)^2 - 4(a + c)(b + c) = -(ab + ac + bc).$$

The usual quadratic criterion over a field of odd characteristic proves the assertion about roots.

In characteristic 3, a zero-sum triple with two equal points has all three points equal. Equality in (7) forces $t = 1$ and $a = b = c$. Conversely, when $a = b = c$, the polynomial in (8) is $2a(t - 1)^2$, whose only root gives precisely this repeated triple. □

The next lemma disposes of all coefficient multiplicities other than three distinct labels. Its proof uses the relation $81 \equiv -1 \pmod{41}$.

Lemma 2.2 (Compatibility of every pair)**.** *For distinct $a, b \in H$, the set $\mathcal{P}(\{a, b\})$ is a cap.*

Proof. Let $r \in H \setminus \{1\}$. Then $r^{81} = r^{-1}$, and Frobenius gives

$$(r - 1)^{80} = \frac{r^{81} - 1}{r - 1} = -r^{-1}, \qquad (r - 1)^{3280} = (-r^{-1})^{41} = -1. \tag{9}$$

Since $3280 = (|F| - 1)/2$, Euler's criterion shows that $r - 1$ is a nonsquare. Every member of H is a square, since $r = (r^{21})^2$, and -1 is a square because $4 \mid |F| - 1$.

The two nonconstant coefficient multisets from $\{a, b\}$ have discriminants $a(b - a)$ and $b(a - b)$. With $r = b/a$, these become

$$a^2(r - 1), \qquad (-r)a^2(r - 1).$$

Both are nonsquares by (9). Lemma 2.1 excludes a zero-sum triple in either case. The all-equal case gives only a repeated point, completing the proof. □

Lemma 2.3 (Opposite-character cosets cannot mix)**.** *If $a \in H$ and $b \in gH$, then $\mathcal{P}(\{a, b\})$ is not a cap.*

Proof. Put $r = b/a$. Since g is primitive, r is a nonsquare; in particular, $r \neq 1$. The two repeated-label multisets have discriminants

$$\Delta(a, a, b) = a(b - a) = a^2(r - 1), \qquad \Delta(a, b, b) = b(a - b) = -a^2 r(r - 1).$$

They are nonzero, and their ratio is $-r$. As -1 is a square, $-r$ is a nonsquare. Exactly one discriminant is therefore a square. Lemma 2.1 applies to that multiset: every coefficient pair in D is nonantipodal, so its quadratic has nonzero leading coefficient and neither excluded ratio 0 nor -1 is a root. The resulting zero-sum triple consists of distinct points. Indeed, in characteristic 3 a zero-sum triple with two equal points has all three equal, while these coefficient labels are not all equal. □

Identify H with $\mathbb{Z}/41\mathbb{Z}$ by $i \mapsto h^i$. Let $\mathcal{G}$ be the 3-uniform hypergraph on this vertex set whose edges are the unordered triples of distinct vertices satisfying

$$\{i, j, k\} \in E(\mathcal{G}) \quad \Longleftrightarrow \quad -(h^{i+j} + h^{i+k} + h^{j+k}) \text{ is a square in } F. \tag{10}$$

For $T \subseteq \mathbb{Z}/41\mathbb{Z}$, write $S(T) = \{h^i : i \in T\}$.

Proposition 2.4. *The set $\mathcal{P}(S(T))$ is a cap if and only if T is independent in $\mathcal{G}$.*

Proof. Lemma 2.2 and the all-equal case in Lemma 2.1 exclude obstructions with at most two coefficient values. For three distinct labels, the normalized quadratic has a root exactly when the corresponding exponent triple is an edge. Such a root gives three distinct points: equality of points on two punctured parabolas would force equality of their coefficients. Conversely, every distinct zero-sum triple with three coefficient values normalizes to such a root. These alternatives exhaust all triples. □

3 Symmetries and a reciprocal obstruction

Translation of exponents and Frobenius preserve $\mathcal{G}$. Indeed, for $u \in H$,

$$\Delta(ua, ub, uc) = u^2\Delta(a, b, c), \qquad \Delta(a^3, b^3, c^3) = \Delta(a, b, c)^3,$$

and both transformations preserve square status in both directions. The corresponding maps on F^2 are $(x, y) \mapsto (x, uy)$ and $(x, y) \mapsto (x^3, y^3)$, which are invertible $\mathbb{F}_3$-linear maps. Thus

$$\Gamma = \{\, i \mapsto 3^j i + t : t \in \mathbb{Z}/41\mathbb{Z},\ 0 \le j < 8 \,\} \tag{11}$$

acts on the independent sets. The order of 3 modulo 41 is 8: $3^4 \equiv -1$ and $3^8 \equiv 1$. Its multipliers are

$$1, 3, 9, 27, 40, 38, 32, 14.$$

In particular, $|\Gamma| = 328$. No other exponent multiplier is needed. Every nonempty independent set can be translated to contain 0; we do not normalize any second vertex.

Lemma 3.1 (Reciprocal obstruction)**.** *For $u \in H$ and $r \in H \setminus \{1\}$, the three coefficients ur^{-1}, u, ur cannot all belong to a set S for which $\mathcal{P}(S)$ is a cap. Equivalently, every nonconstant three-term arithmetic progression in $\mathbb{Z}/41\mathbb{Z}$ is an edge of $\mathcal{G}$.*

Proof. The coefficients $1, r, r^{-1}$ are distinct, since H has odd prime order. In characteristic 3,

$$\Delta(1, r, r^{-1}) = -(r + 1 + r^{-1}) = -\frac{(r-1)^2}{r}. \tag{12}$$

This is a nonzero square, because r and -1 are squares. Proposition 2.4 supplies the obstruction. Scaling by u preserves it, and writing $u = h^i$, $r = h^d$ gives the exponent progression $\{i - d, i, i + d\}$. □

4 The finite certificate and the exact maximum

We use the ordinary representatives $0, 1, \ldots, 40$ for the vertices when ordering sets. The complete hypergraph has

$$\binom{41}{3} = 10\,660$$

potential edges. Computing (10) on every triple gives 5740 edges and 4920 nonedges. None of the 10 660 discriminants is zero. The degree of each vertex is 420, consistently with translation symmetry and $3 \cdot 5740 = 41 \cdot 420$.

The second component of the certificate is a family $\mathcal{C}$ of subsets containing 0. It is supplied as a complete list, with each set recorded in increasing order. Its cardinality distribution is

Set size	1	2	3	4	5
Number in $\mathcal{C}$	1	40	360	488	40

Table 1: The independent sets containing 0. The certificate has 929 entries and no entries of another size.

The fact needed for the proof is completeness of this list. The following elementary criterion separates that issue from the algorithm that produced the data.

Lemma 4.1 (Increasing extension criterion)**.** *Let $\mathcal{K}$ be a hypergraph on $\{0, 1, \dots, n-1\}$ and let $\mathcal{A}$ be a finite family of vertex sets. Suppose that*

1. *$\{0\} \in \mathcal{A}$;*
2. *every member of $\mathcal{A}$ contains 0 and is independent in $\mathcal{K}$;*
3. *for every $I \in \mathcal{A}$ and every integer v with $\max I < v < n$, if $I \cup \{v\}$ is independent, then $I \cup \{v\} \in \mathcal{A}$.*

Then $\mathcal{A}$ is exactly the family of independent sets containing 0.

Proof. Let $J = \{0, j_2, \dots, j_s\}$ be independent, with $0 < j_2 < \cdots < j_s$. Starting with $\{0\}$, its successive initial segments belong to $\mathcal{A}$ by condition (3), since they are independent subsets of J. Hence $J \in \mathcal{A}$. Condition (2) gives the reverse inclusion. □

Proposition 4.2 (Finite verification)**.** *The supplied hypergraph table agrees with (10) on all 10 660 triples. The supplied family $\mathcal{C}$ satisfies Lemma 4.1 for $\mathcal{K} = \mathcal{G}$, has the counts in Table 1, and contains $\{0, 2, 6, 13, 23\}$.*

This proposition is the computational part of the proof. Its checks use exact arithmetic and exhaustive finite lists. For each listed set, the verifier checks distinctness, increasing order, membership of 0, and the absence of a hyperedge. It then tests every larger possible next vertex, including those extending the largest listed sets. There are 10 182 extension tests, of which 928 give independent children; each such child occurs in the list. Together with the root $\{0\}$, these are the 929 entries. Thus the absence of a six-element entry is supported by extension closure, not merely by a reported search maximum. Section 6 specifies the arithmetic and the reproduction procedure.

Proof of the maximum and attainment in Theorem 1.1. By Proposition 4.2 and Lemma 4.1, every independent set containing 0 is in $\mathcal{C}$, and therefore has size at most 5. Every nonempty independent set has a translate containing 0, so the same bound holds without normalization. The empty set causes no exception. Proposition 2.4 gives $|S| \le 5$ for every admissible $S \subseteq H$.

The exponent set $\{0, 2, 6, 13, 23\}$ belongs to $\mathcal{C}$, so the same proposition proves caphood of $\mathcal{P}(S_0)$. Formula (2) gives $5(6561 - 1) = 32\,800$ points. □

There is also a direct check of this attainment which does not use the hypergraph flags. Among the five coefficients there are $\binom{7}{3} = 35$ multisets of size three: five all-equal, twenty with exactly two equal labels, and ten with three distinct labels. Choose one ordering of each multiset and evaluate (8) for every $t \in F \setminus \{0, -1\}$. These are $35 \cdot 6559 = 229\,565$ equations. The only roots are the five all-equal cases at $t = 1$. Every triple of points can be reordered and then normalized, so this check covers all coefficient orders and first-coordinate choices. The companion supplement describes it and gives a second upper proof using a link graph and 24 obstruction orbits. The two upper certificates use the same hypergraph; they provide different completeness arguments for that fixed relation.

5 Classification of the maximizers

The certificate gives exactly 40 independent five-sets containing 0. Translation is transitive on vertices and preserves independence, so every vertex lies in exactly 40 independent five-sets. If N denotes the number of independent five-sets without normalization, counting pairs (v, T) with $v \in T$ gives

$$5N = 41 \cdot 40, \qquad N = 328. \tag{13}$$

To turn this count into the stated classification, it remains to exclude nontrivial stabilizers in Γ.

Lemma 5.1. *The action of Γ on the independent five-sets of $\mathcal{G}$ is free.*

Proof. Let T be such a five-set, and suppose that the affine map $f(i) = ki + t$, with $k \in \langle 3 \rangle$, stabilizes T. If $k = 1$ and $t \neq 0$, this is a translation with a single orbit of length 41 on the vertex set. It cannot preserve a set of size 5.

Suppose that $k \neq 1$, and let d be its multiplicative order. Then $d \in \{2, 4, 8\}$ and

$$f^d(i) = i + t(1 + k + \cdots + k^{d-1}) = i.$$

The geometric sum vanishes because $k \neq 1$, while no smaller positive power can have multiplier 1. Thus f has order d, and $f^{d/2}$ is a nonidentity involution stabilizing T. Its multiplier is -1, so it has the form

$$i \longmapsto -i + u.$$

This involution has the unique fixed point $u/2$ in $\mathbb{Z}/41\mathbb{Z}$, and all other vertex orbits have size two. Since $|T| = 5$ is odd, T contains $u/2$ and at least one opposite pair about that point. Those three vertices form a nonconstant arithmetic progression, contrary to Lemma 3.1. Consequently the only stabilizer element is the identity. □

Completion of the proof of Theorem 1.1. The orbit of the independent five-set $\{0, 2, 6, 13, 23\}$ has $|\Gamma| = 328$ members by Lemma 5.1. Equation (13) shows that these are all independent five-sets. Under $i \mapsto h^i$, the action $i \mapsto 3^j i + t$ becomes $c \mapsto h^t c^{3^j}$. Proposition 2.4 now gives (5), including the distinctness of the 328 coefficient sets. □

Proof of Theorem 1.2. By Lemma 2.3, an admissible coefficient set $S \subseteq D$ is contained wholly in H or wholly in gH. Multiplication of all coefficients by g is induced by the invertible $\mathbb{F}_3$-linear map $(x, y) \mapsto (x, gy)$, so the H-classification in Theorem 1.1 transfers to gH. Each coset therefore has maximum five and exactly 328 maximizing coefficient sets. Since the cosets are disjoint, these give precisely 656 distinct sets in D, with the two displayed descriptions. Equation (2) gives the point count. □

6 Exact arithmetic and reproducibility

The proof has a small algebraic part and a finite exhaustive part. The lemmas above establish the correspondence with caphood, the valid normalization, and the classification from the rooted count. The numerical assertions in Proposition 4.2, including the particular field

coordinates, are verified by the supplied program and data. We describe their content so that the logical role of each check is explicit.

Represent an element $\sum_{i=0}^{7} a_i z^i$ by the integer

$$\mathrm{id}\left(\sum_{i=0}^{7} a_i z^i\right) = \sum_{i=0}^{7} a_i 3^i, \qquad a_i \in \{0,1,2\}. \tag{14}$$

These integers are labels; their ordinary integer operations are not field operations. Addition is coefficientwise modulo 3, and multiplication uses polynomial reduction by $z^8 = 1 + 2z^2$. The labels of g and h are 38 and 993. The powers of g form a cycle of length 6560 containing every nonzero residue of the quotient in (3). This check proves at once that every nonzero residue is a unit, that the quotient is a field, and that g is primitive. The identity $g^{160} = h$ and the 41-element cycle of h verify the subgroup coordinates. The labels of the witness coefficients, in the order of (4), are

$$1, \quad 1063, \quad 3121, \quad 5401, \quad 4873.$$

The hypergraph data record, for every $0 \le i < j < k \le 40$, the exact label of $\Delta(h^i, h^j, h^k)$ and its square flag. The verifier checks that every triple occurs exactly once, recomputes its discriminant in the quotient, and checks square status. Nonzero squares can be identified as the even powers of the verified primitive element; zero is treated according to the convention in Section 2. The rooted family data contain the 929 explicit increasing lists. Triple lookups give validity and every extension obligation in Lemma 4.1. Neither the proof of coverage nor its checking requires importing the program that originally generated this family.

The portable companion archive contains these data, a standard-library Python verifier, the witness, the supplementary link certificate, and instructions for reproducing the checks. From the archive root, its entry point is `verification/verify.py`; the accompanying README gives the command and receipt options for the delivered version. The verifier performs the field and subgroup checks, all discriminant checks, complete family closure, and direct witness validation. Its outputs are verification receipts, separate from the mathematical inputs. Source hashes identify the supplied files; the explicit checks, rather than the hashes, support the finite assertions.

The same archive supplies `verification/verify_two_cosets.py`. It independently implements the field arithmetic, checks primitivity of g and the quadratic characters of both cosets, and recomputes both repeated-label discriminants for all $41^2 = 1681$ cross-coset pairs. It also enumerates the $\binom{82}{3} = 88\,560$ distinct-label discriminants as a diagnostic; this enumeration is not needed for Theorem 1.2, since a mixed pair is already forbidden. The symbolic proof of Lemma 2.3 and the checked H-certificate are the two necessary parts of the extended classification.

The companion supplement records the link-graph obstruction table and the direct normalized-equation test. Those checks provide further verification of the upper bound and of attainment, while the complete rooted family supplies the count used for classification.

References

[1] C. Grace. capsets: code accompanying algebraic capsets. `https://github.com/cassie-grace/capsets`, 2026. Revision `d74441822bcf`, 5 February 2026.

[2] C. Grace and J. F. Voloch. Algebraic capsets. *Journal of Combinatorial Designs*, 34(10):459–463, 2026. Publisher version. https://doi.org/10.1002/jcd.70024.